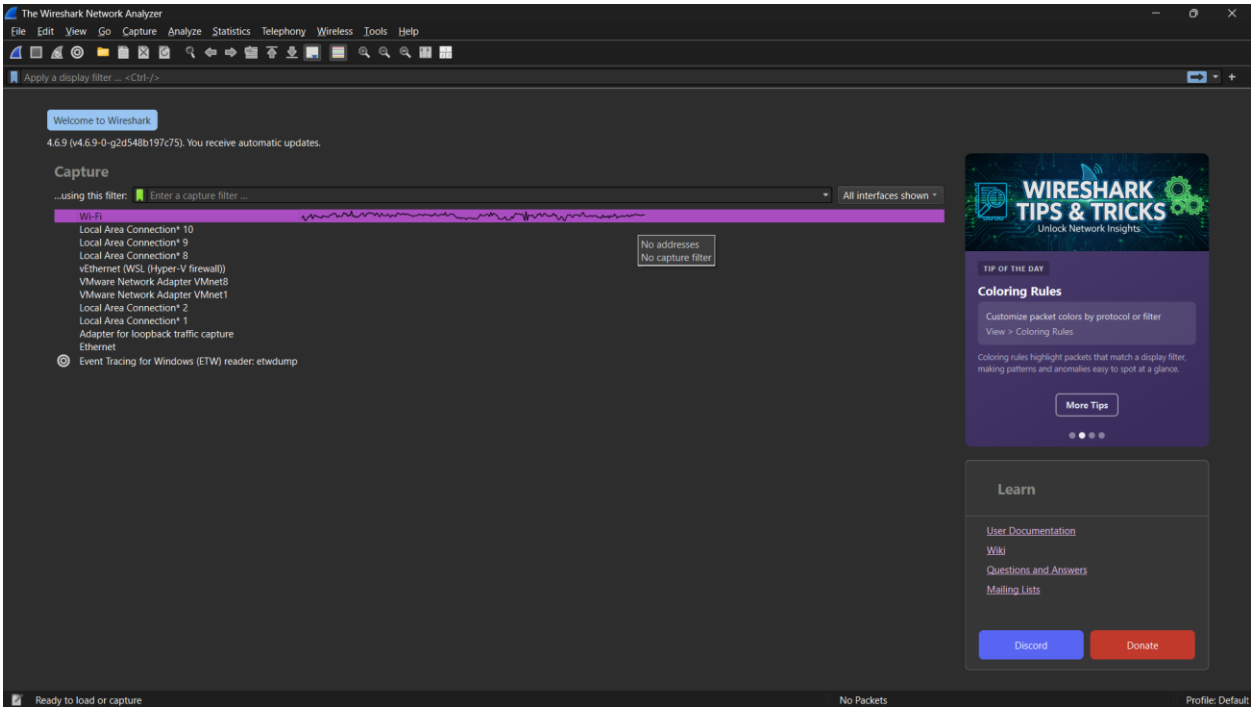
CNNP Skill-1 Lab Experiment

Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission. Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead.

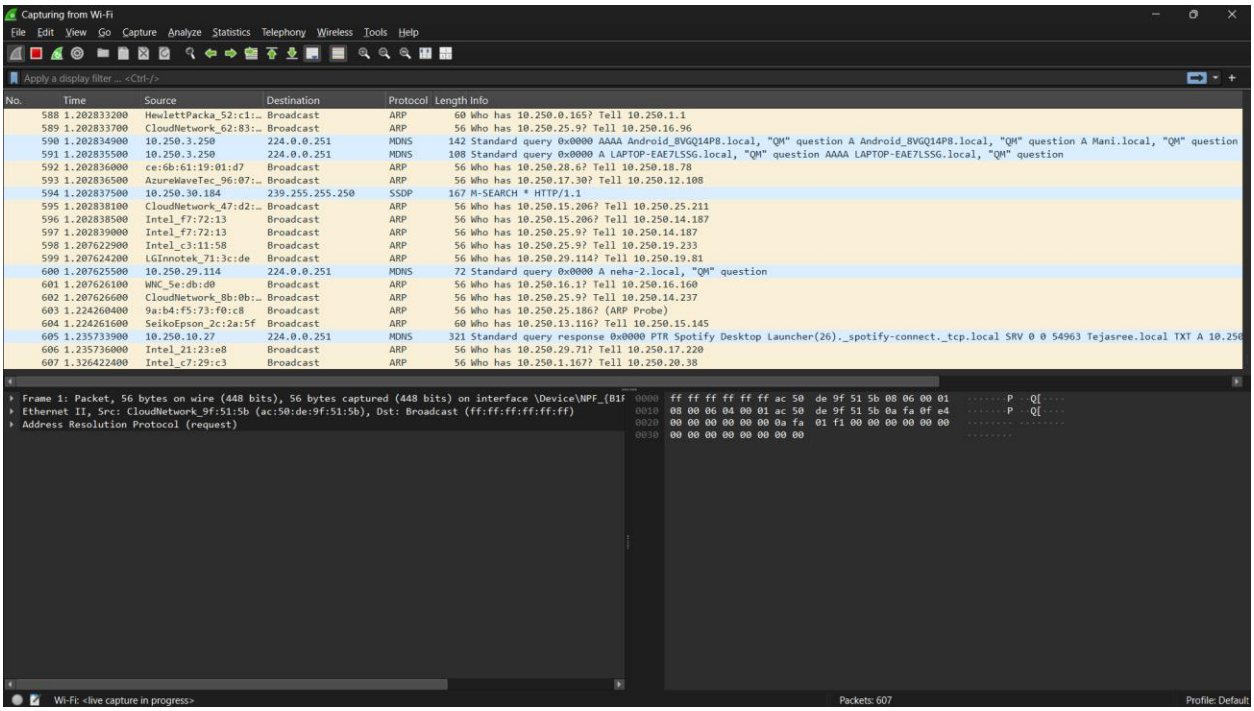
Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Stage 1 – Open Wireshark



Stage 2 – Select your network interface

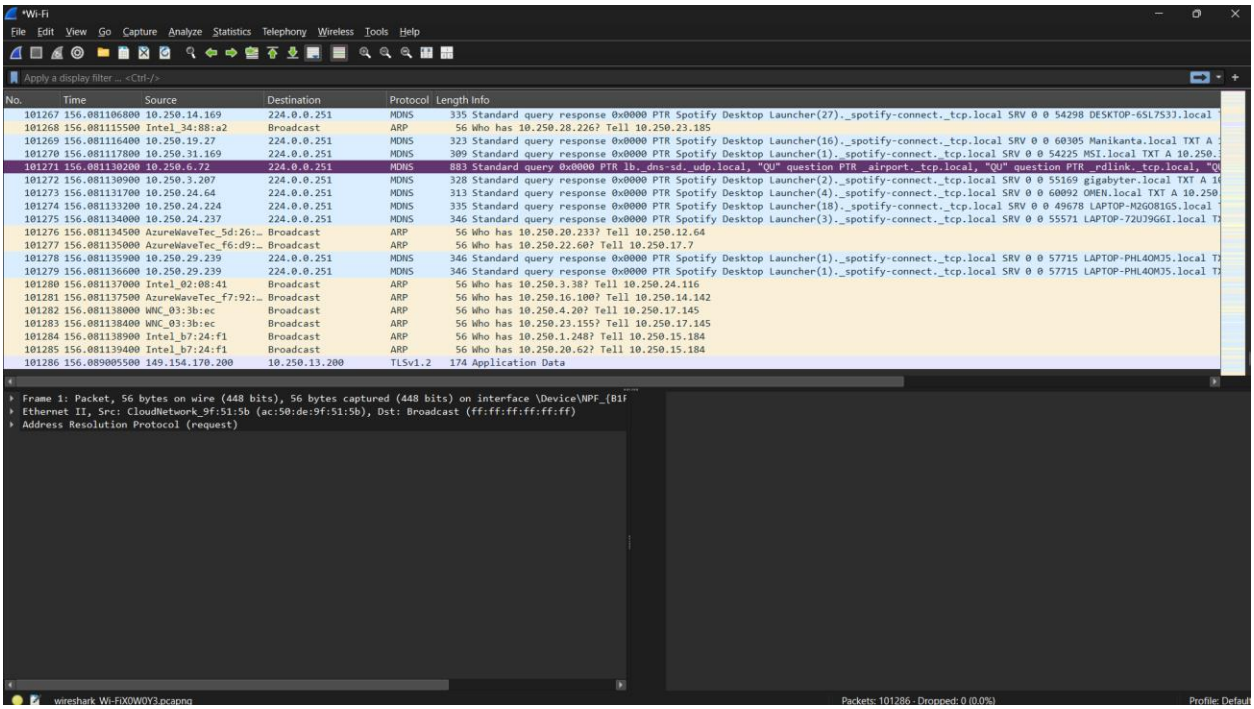
In this case we are using WIFI that is connected to internet so we click on WIFI and open it.



Stage 3 – Generate network traffic

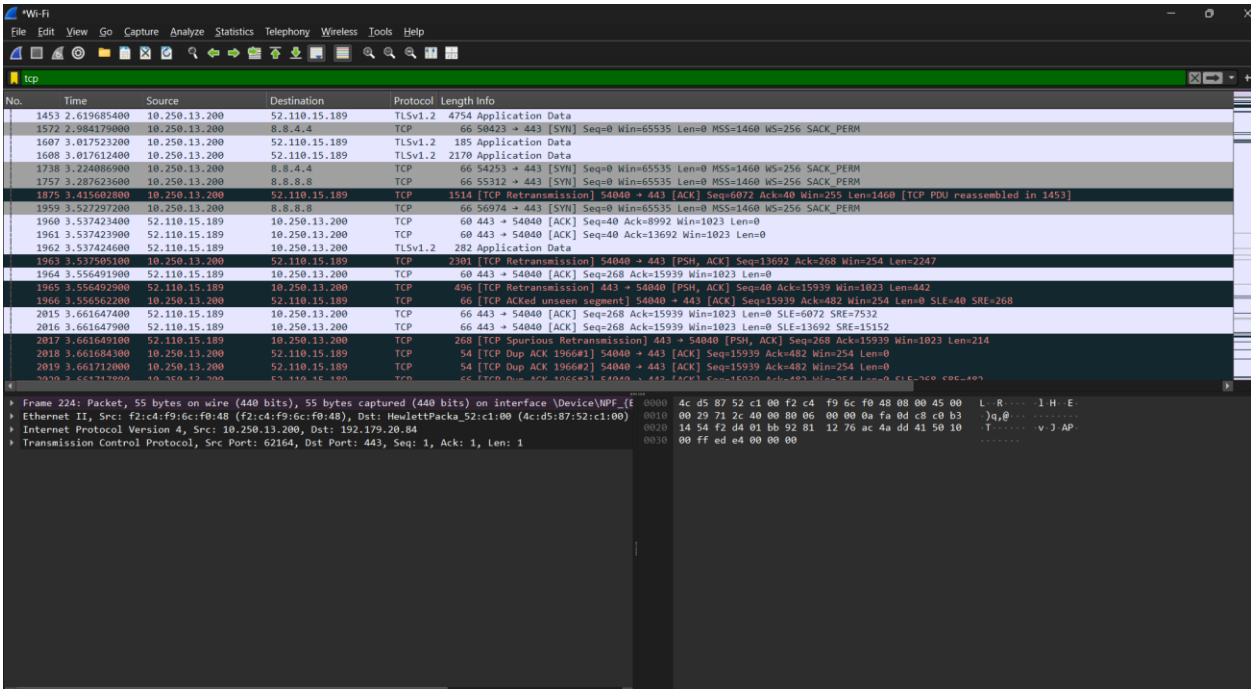
While Wireshark is capturing:

1. Open your web browser.
2. Visit https://www.google.com.
3. Open two or three other websites.
4. Wait approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button at the top.

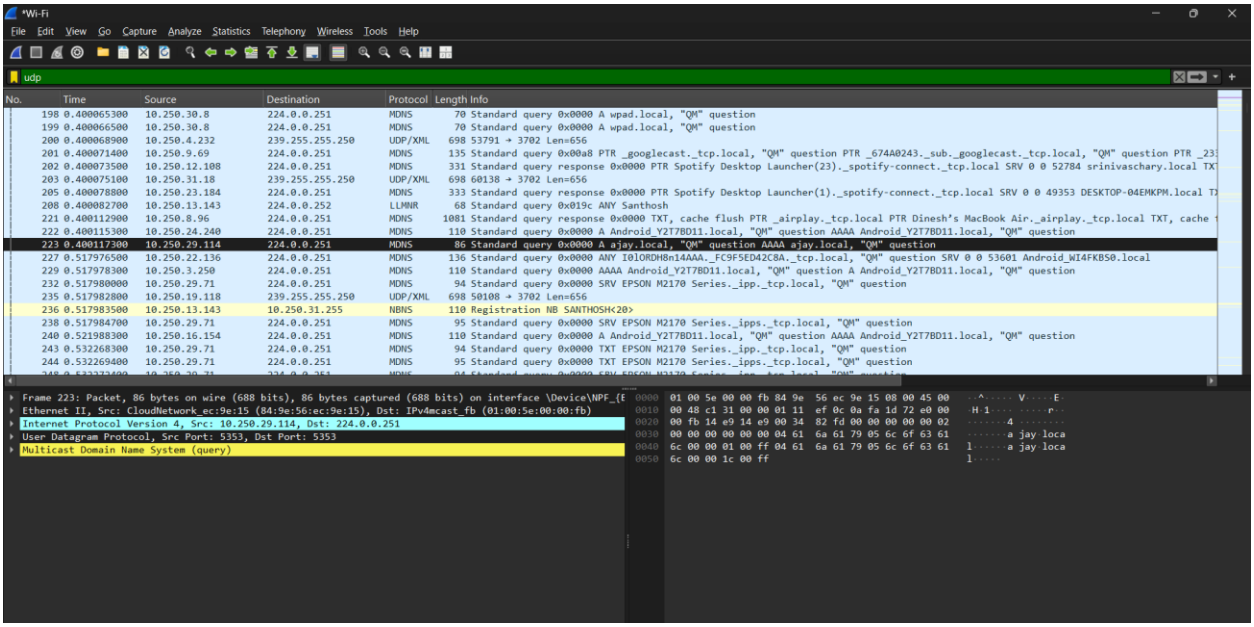


Stage 4 – Apply display filters

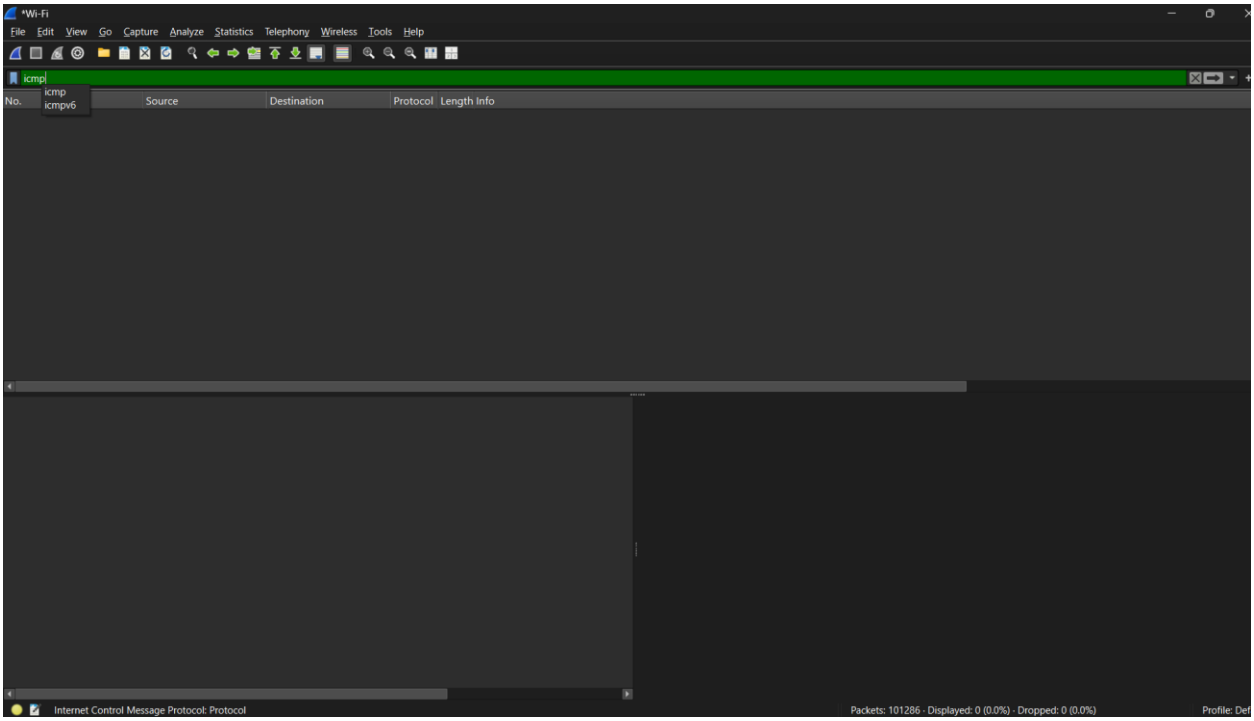
1. TCP filter



2. UDP filter

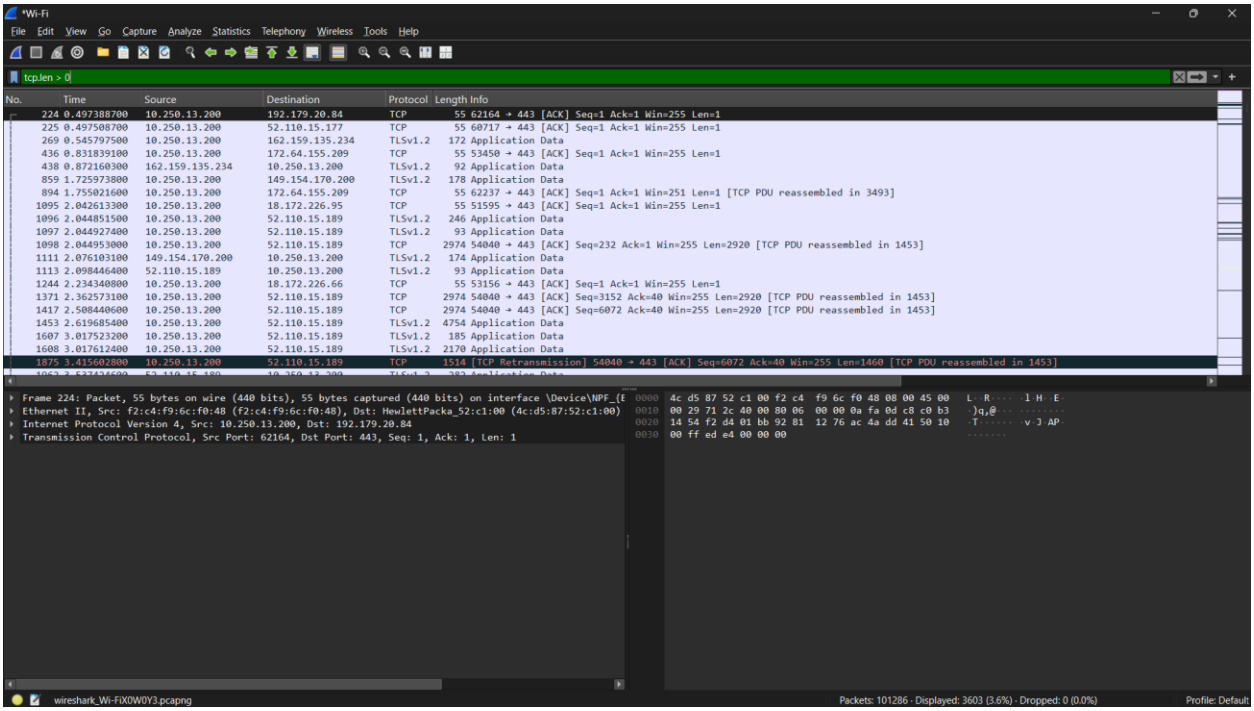


3. ICMP filter



4. TCP data filter

tcp.len > 0



Stage 5 – Inspect packet headers

Parameter	TCP Packet	UDP Packet
Frame Length	66 bytes	60 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	32 bytes	8 bytes
IP Total Length	52 bytes	46 bytes

Payload Length	12 bytes	18 bytes

Stage 6 – Calculate Header Overhead

Component	Size
Ethernet Header	14 bytes
IPv4 Header	20 bytes
TCP Header	20 bytes
Application Data	1000 bytes
FCS	4 bytes

Total size including FCS :-

14+20+20+1000+4 = 1058

Total overhead :-

14+20+20+4 = 58

Payload efficiency :-

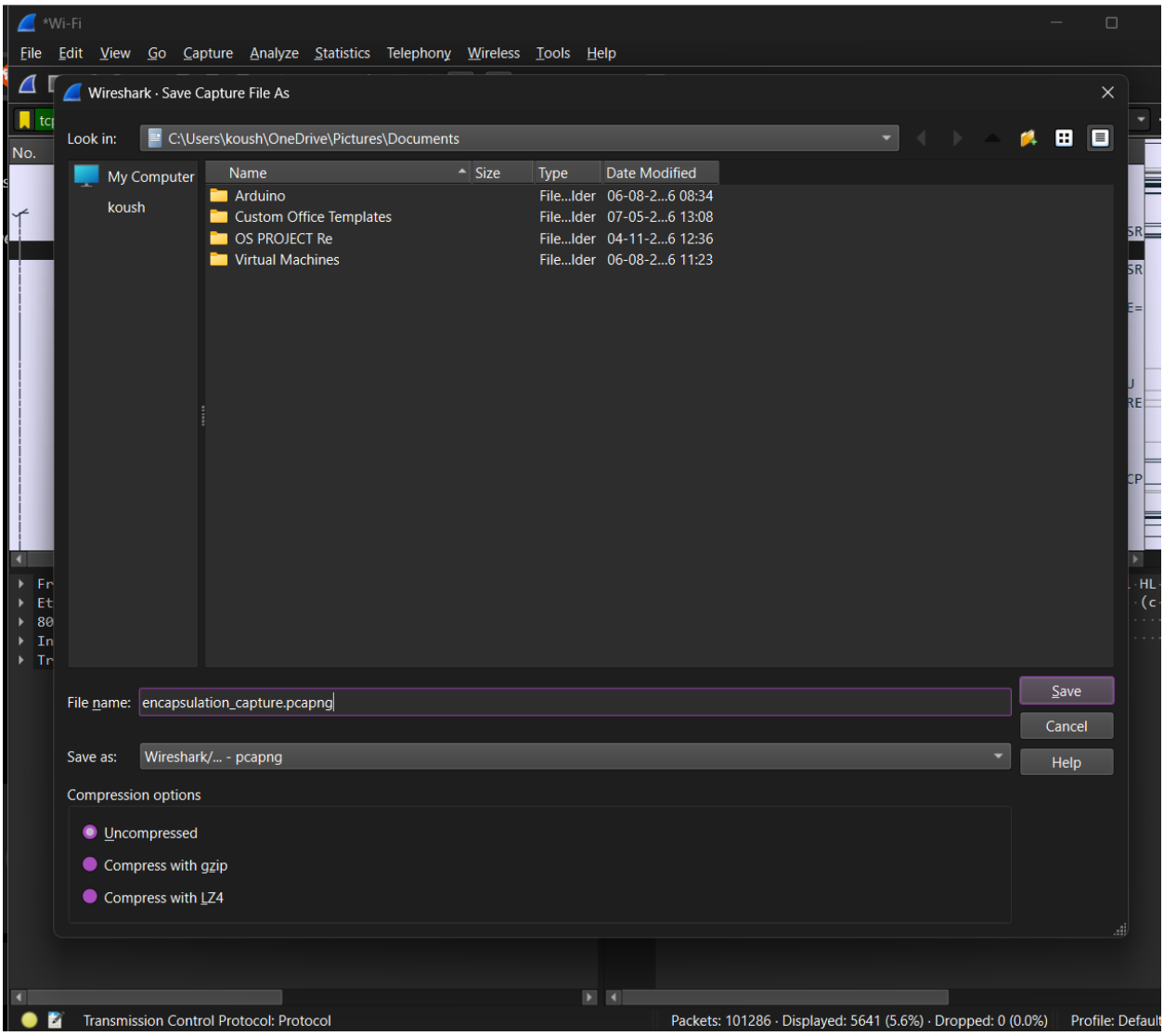
1000/1058 * 100 = 94.52%

Stage 7 – Observation Table

Parameter	TCP Packet	UDP Packet
Frame Length	66 bytes	60 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	32 bytes	8 bytes
IP Total Length	52 bytes	46 bytes
Payload Length	12 bytes	18 bytes

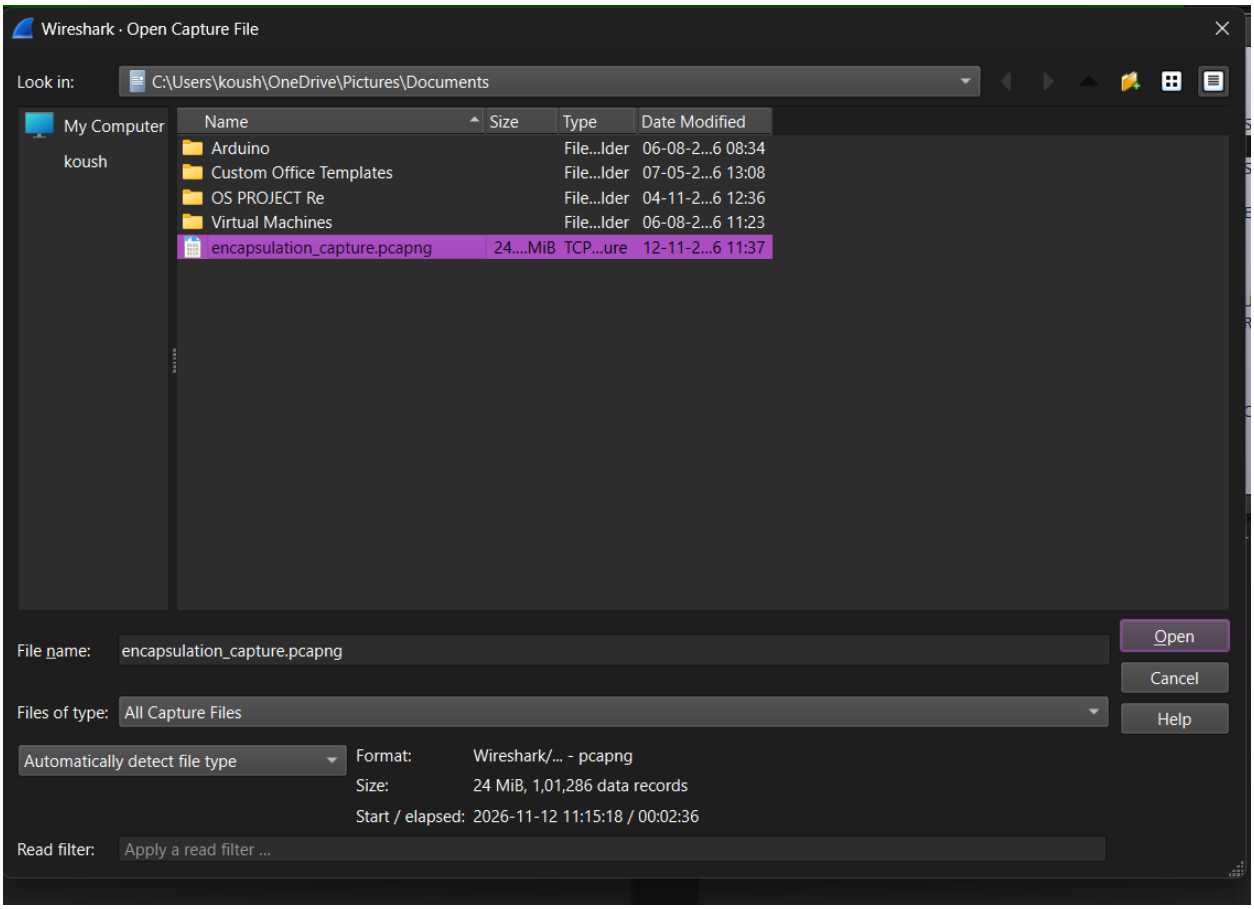
Stage 8 – Save the Capture File

1. Open Wireshark.
2. Click File → Save As.
3. Select Documents.
4. Enter the filename `encapsulation_capture.pcapng`.
5. Keep the format as pcapng.
6. Click Save.



Stage 9 – Open the Saved Capture

1. Click File → Open.
2. Select `encapsulation_capture.pcapng`.
3. Click Open.
4. Type `icmp` in the display filter bar.
5. Press Enter.



Stage 10 – Answer the Questions

1. What is encapsulation?

Encapsulation is the process of adding protocol headers to data at each network layer before transmission.

2. What is the purpose of the Ethernet header?

It contains source and destination MAC addresses and identifies the type of data being carried.

3. What information is stored in the IP header?

It contains source and destination IP addresses, packet length, TTL and protocol information.

4. What is the difference between TCP and UDP headers?

TCP has a larger header with sequence numbers, acknowledgment numbers and control flags. UDP has a simple 8-byte header.

5. Why does each protocol add a header?

Headers provide information needed for addressing, routing and proper data delivery.

6. What is packet overhead?

Packet overhead is the additional information added to application data by network protocols.

7. How does header overhead affect network efficiency?

Higher header overhead reduces the percentage of a packet available for actual application data.